

Do Positive Reviews Influence Audience Interest?

Global Box Office Performance (2000–2024)

TECHNOVA

IT3011 - Theory and Practices in Statistical Modelling

The slide features decorative blue geometric shapes in the corners, consisting of overlapping triangles and polygons in various shades of blue.

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Introduction

Understanding how feedback affects audience behavior is important in today's decision-making environment. People often rely on reviews and feedback when choosing what to engage with. Higher positive feedback can attract more attention and participation, while lower feedback may reduce interest. This study aims to examine whether such feedback has a meaningful impact on audience interest.

The image features a white background with four decorative corner elements. Each corner contains overlapping triangles and trapezoids in various shades of blue and purple. The top-left and bottom-right corners have more complex, multi-layered geometric patterns, while the top-right and bottom-left corners feature simpler, more linear shapes.

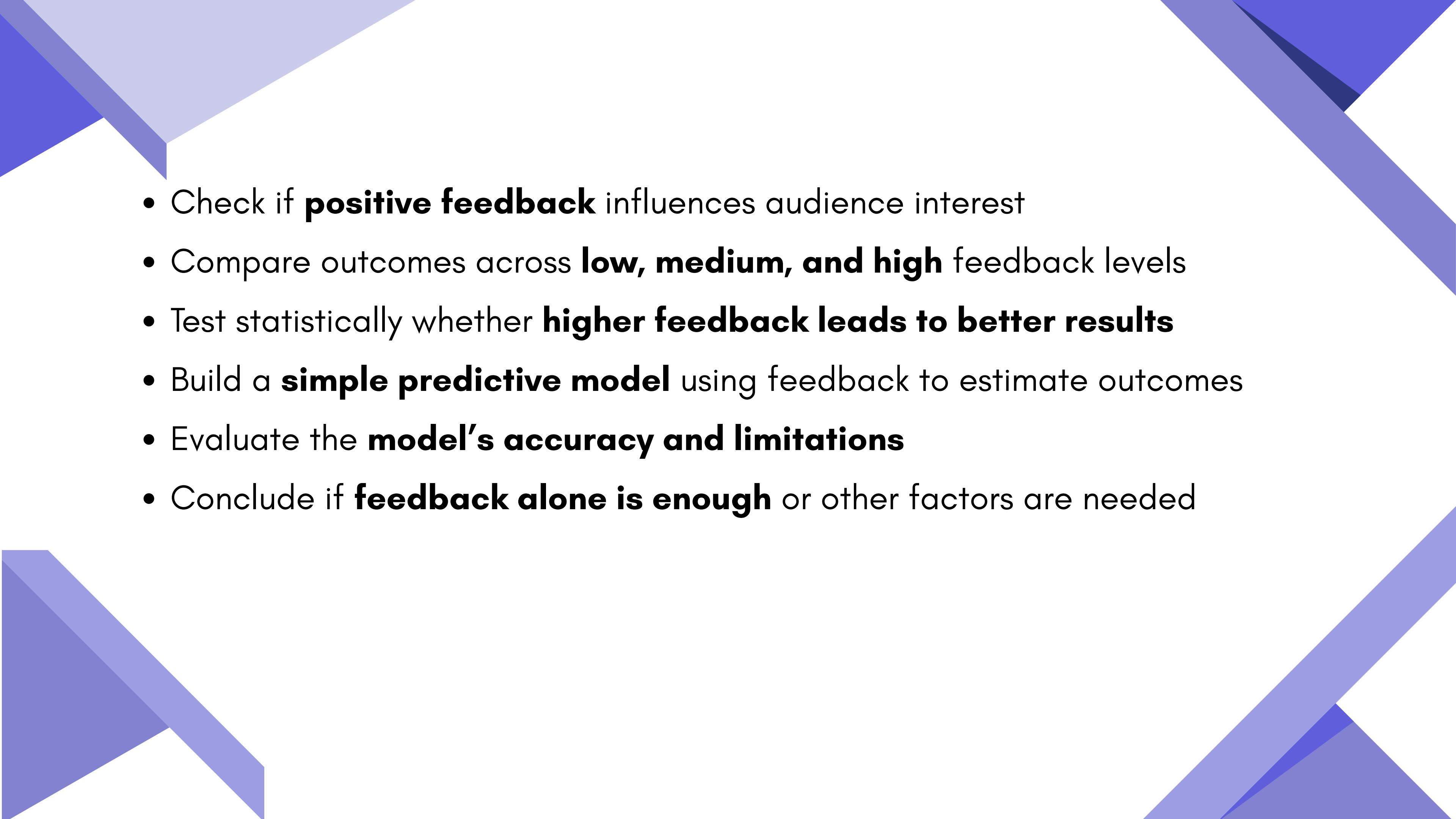
problem statement

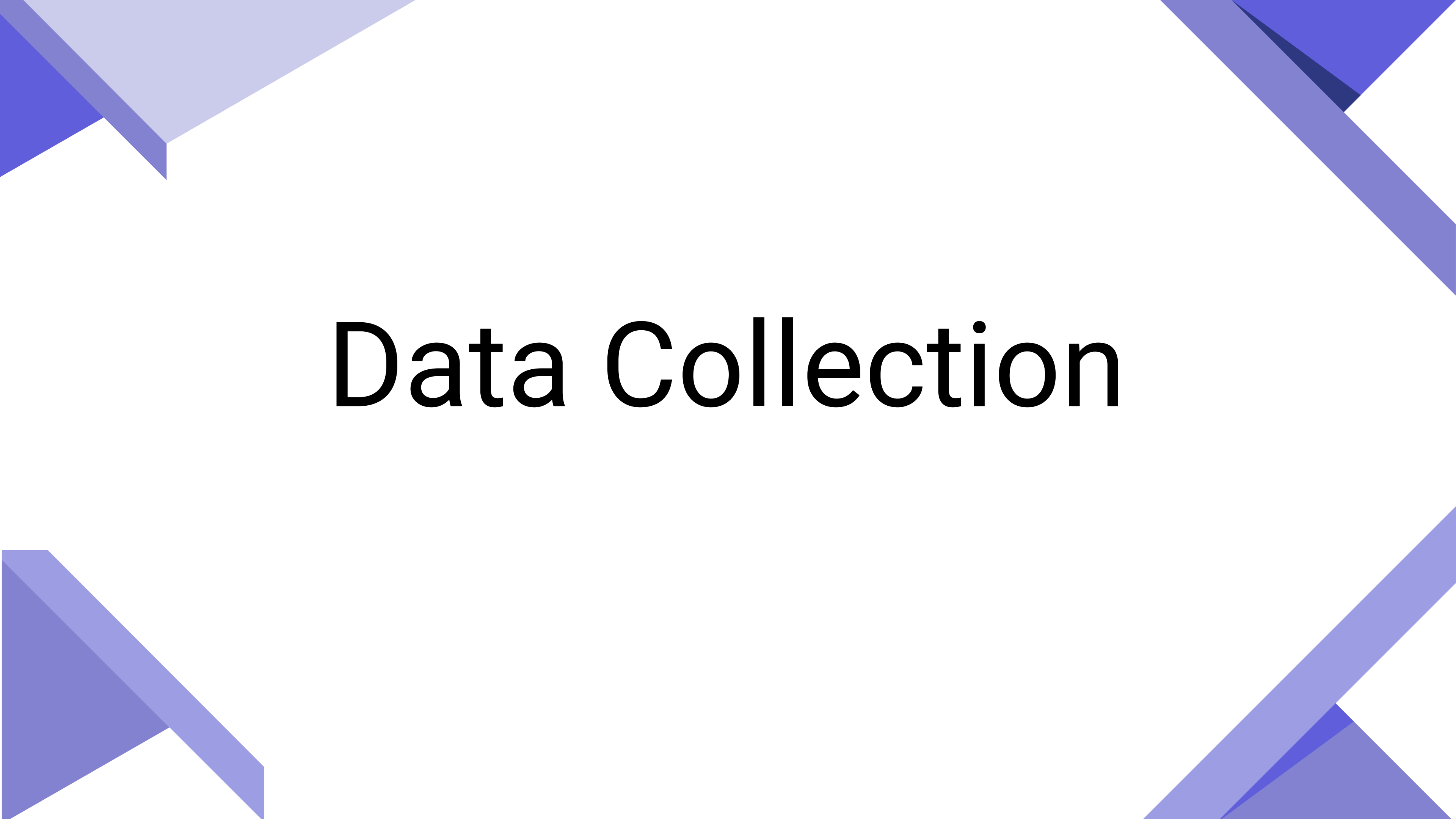
Do Positive Reviews Influence Audience Interest?

- Investigate how **positive feedback** affects audience interest
- Examine if there is a **clear link between feedback and success**
- Determine whether **feedback alone can predict audience behavior**
- Provide insights for **decision-makers and stakeholders**



objectives

- 
- Check if **positive feedback** influences audience interest
 - Compare outcomes across **low, medium, and high** feedback levels
 - Test statistically whether **higher feedback leads to better results**
 - Build a **simple predictive model** using feedback to estimate outcomes
 - Evaluate the **model's accuracy and limitations**
 - Conclude if **feedback alone is enough** or other factors are needed

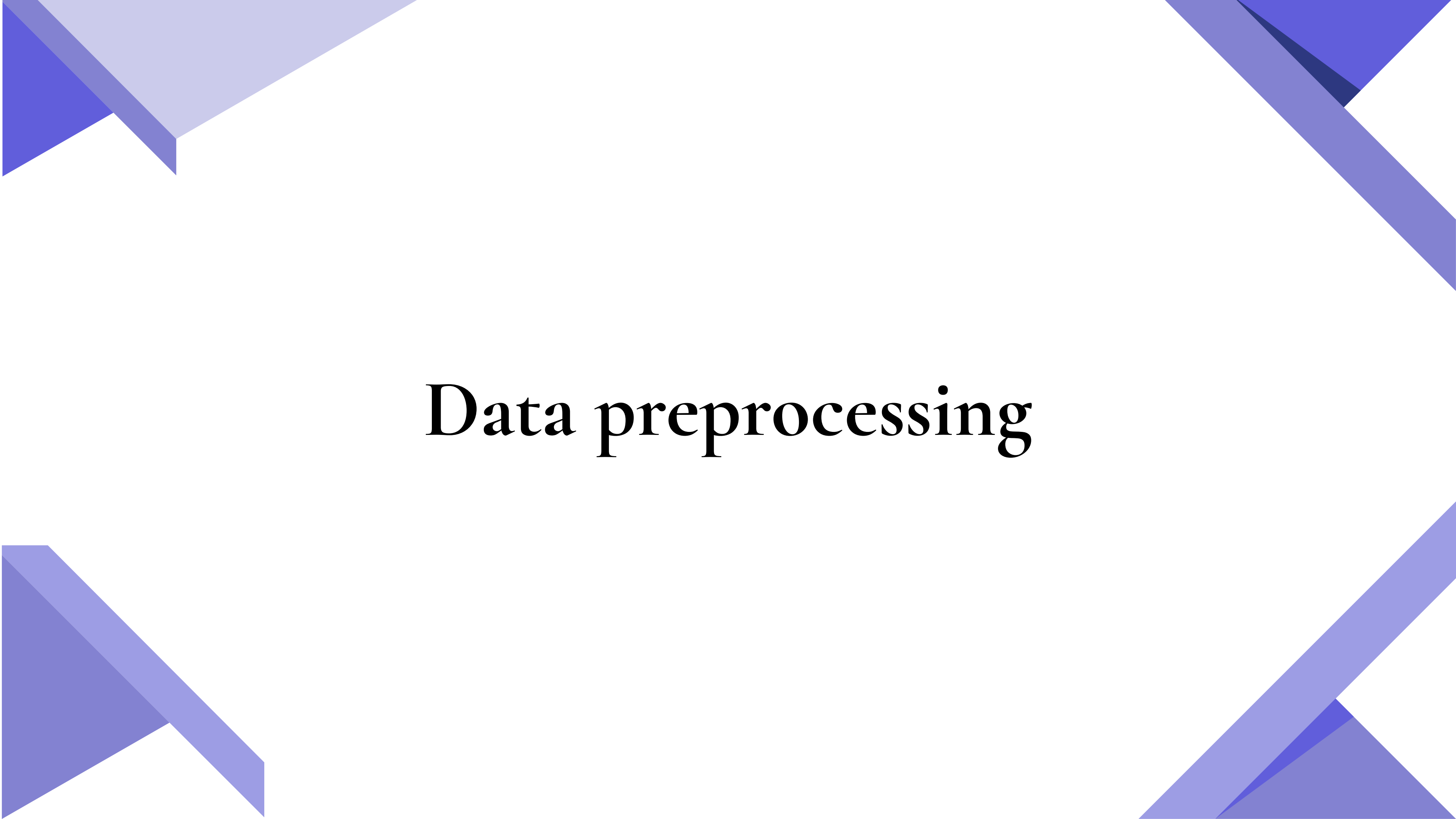


Data Collection

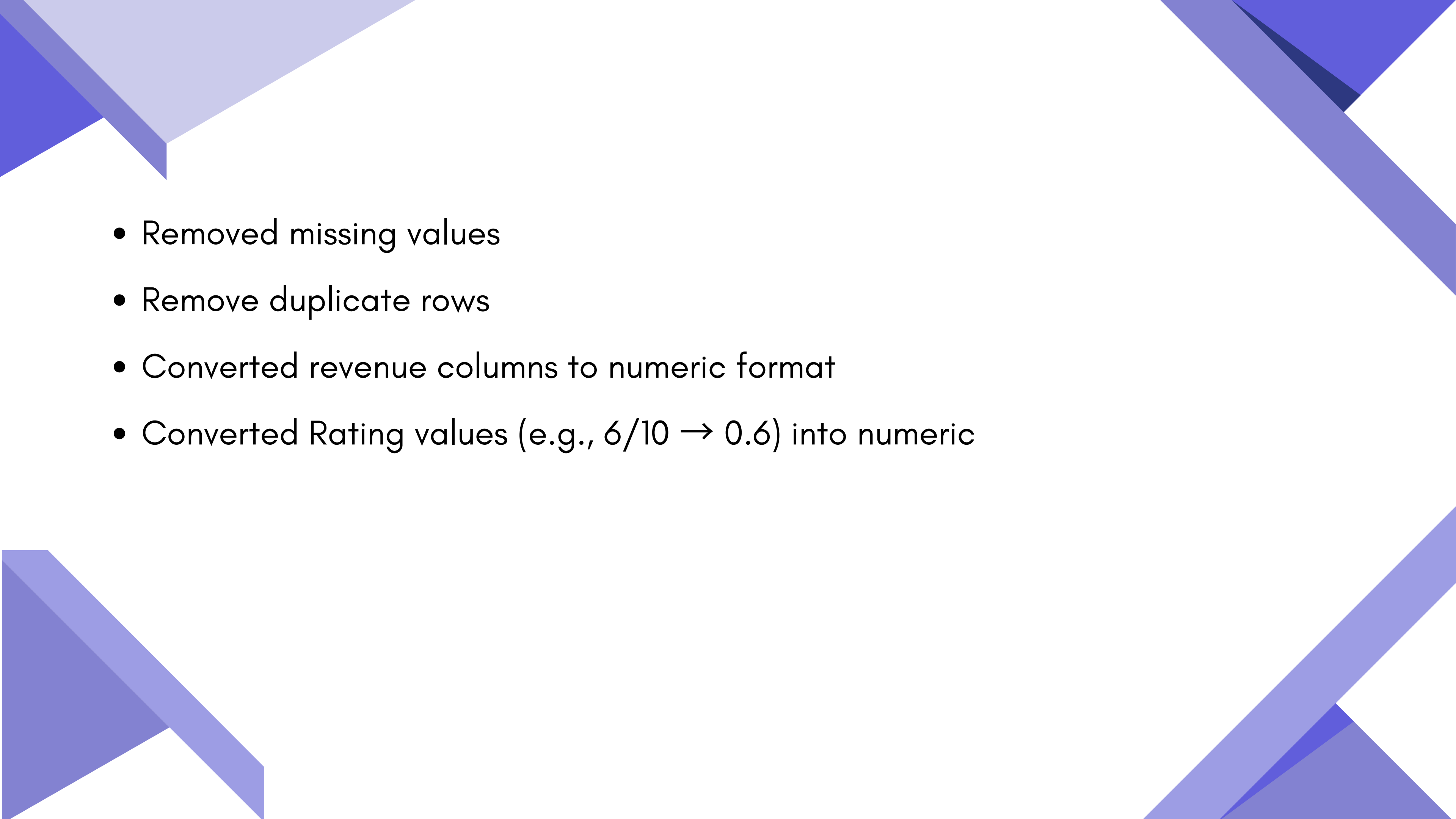


Kaggle Dataset: Movies Box office Dataset (2000-2024)

- Dataset covers global performance from 2000-2024.
- Focused on audience feedback scores and overall earnings.
- Audience feedback represents positive reviews, showing how viewers perceive content.
- Overall earnings represent audience interest, reflecting viewer engagement.
- Helps analyze how ratings guide decisions and whether higher feedback attracts more interest.
- Low feedback may indicate reduced interest, even if the content is widely produced.



Data preprocessing

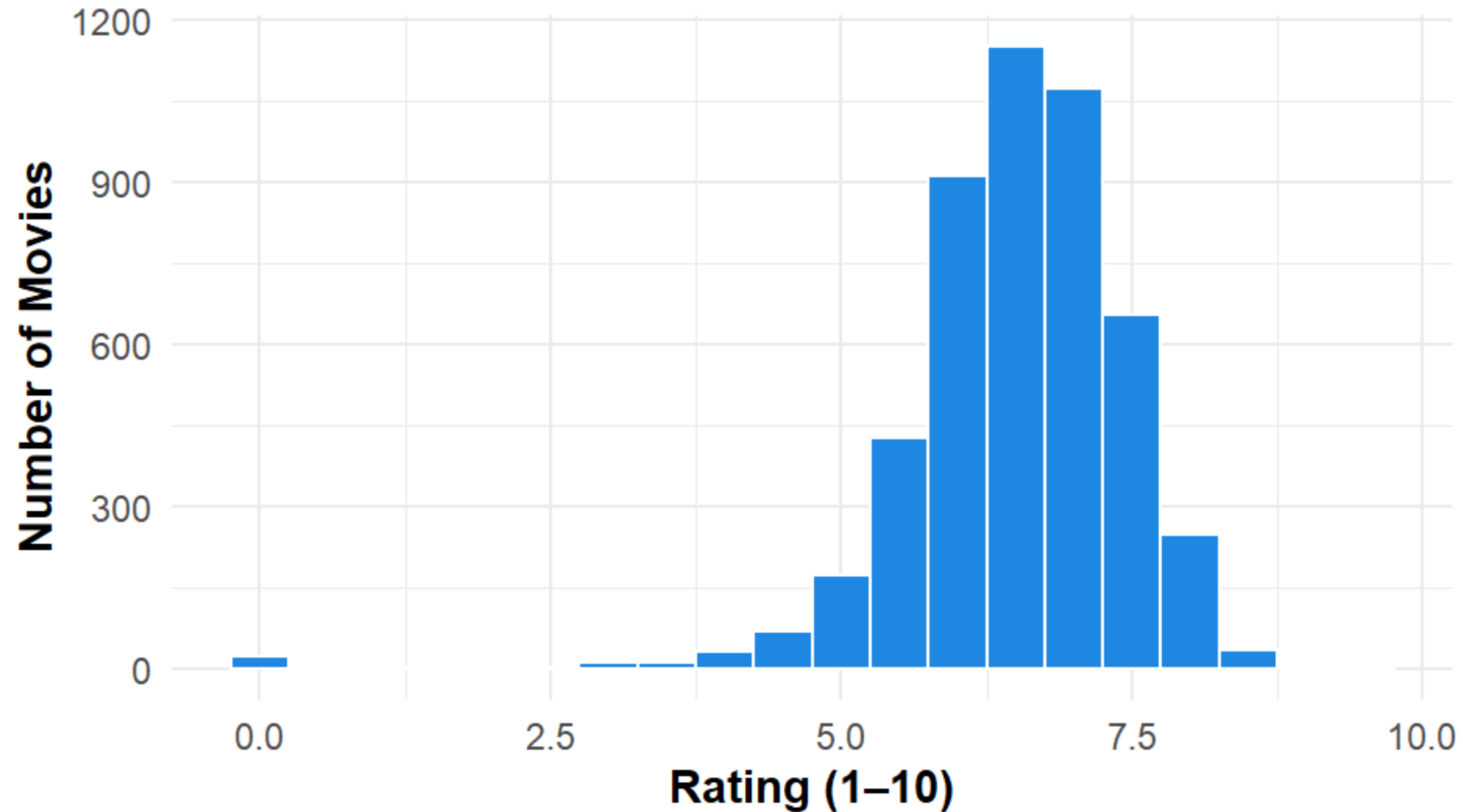
- 
- Removed missing values
 - Remove duplicate rows
 - Converted revenue columns to numeric format
 - Converted Rating values (e.g., 6/10 $\rightarrow$ 0.6) into numeric



Descriptive Analysis

understand the basic pattern of the data

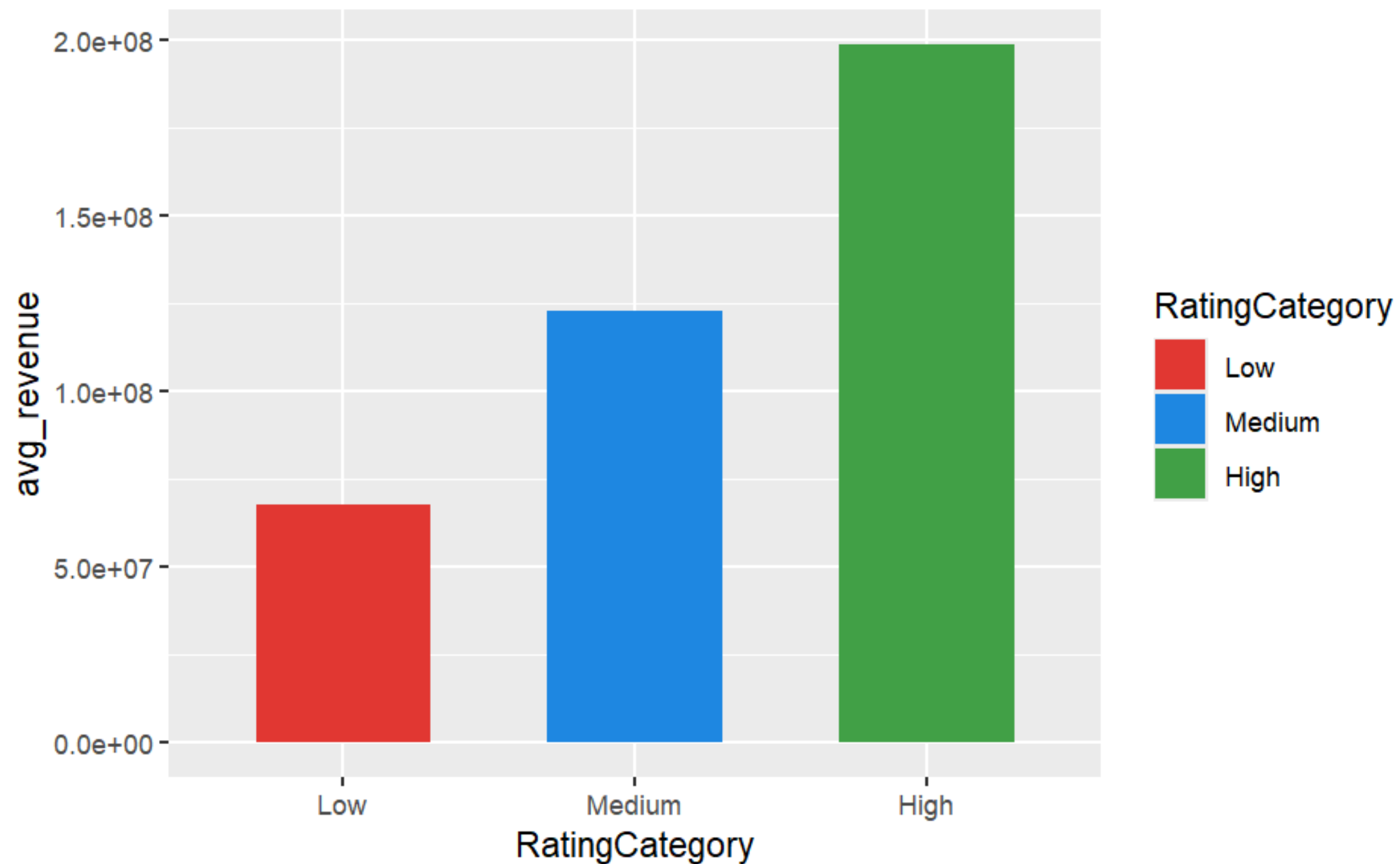
Distribution Of Movie Ratings



Low Ratings ≤ 5.9
Medium Ratings 6.0 – 7.4
High Ratings ≥ 7.5

- Most feedback scores are concentrated in the medium range.

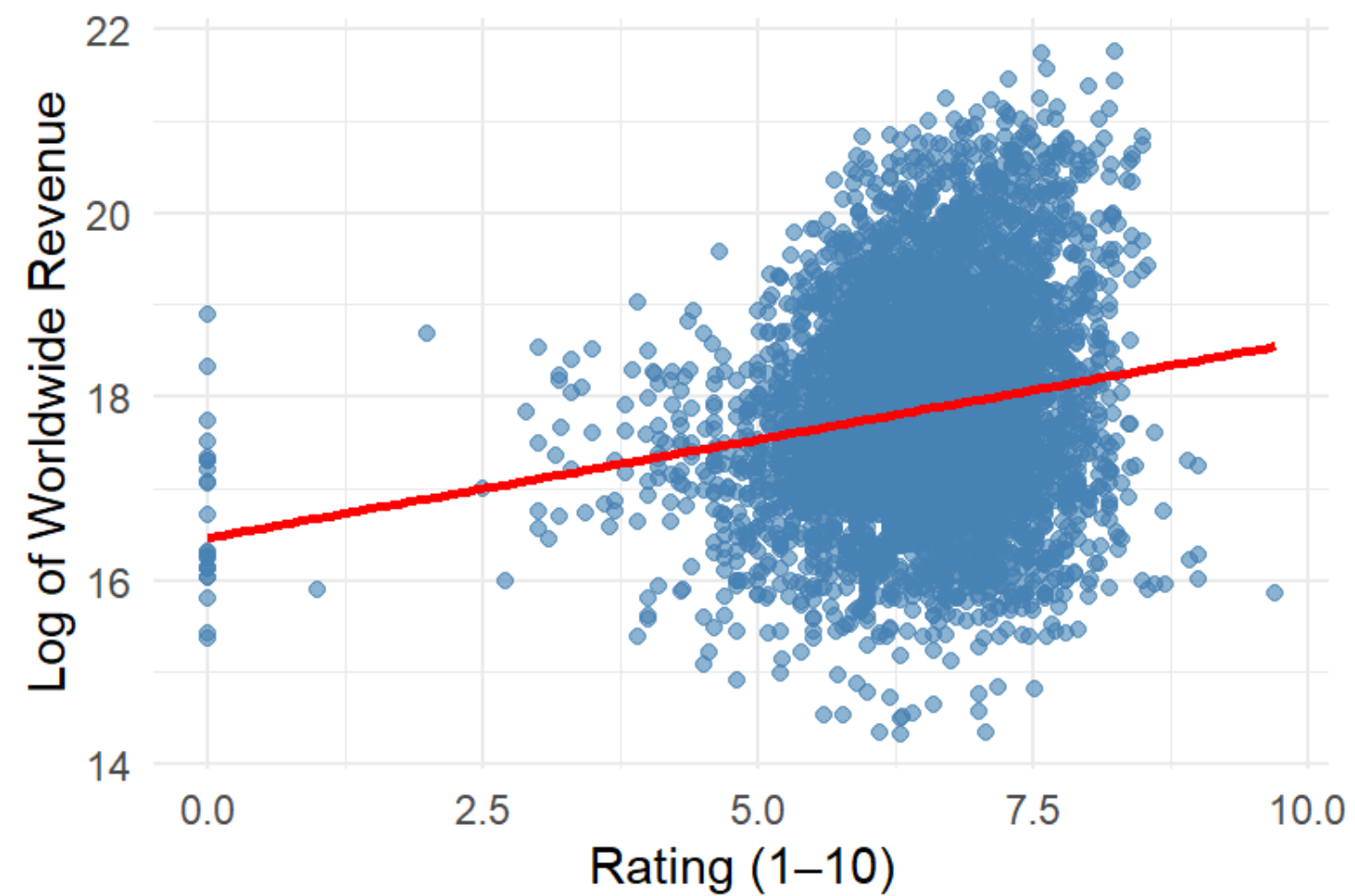
Average Revenue by Rating Category



Low Ratings ≤ 5.9
Medium Ratings 6.0 – 7.4
High Ratings ≥ 7.5

- Performance increases as feedback increases.

Scatter Plot : Rating vs LogRevenue



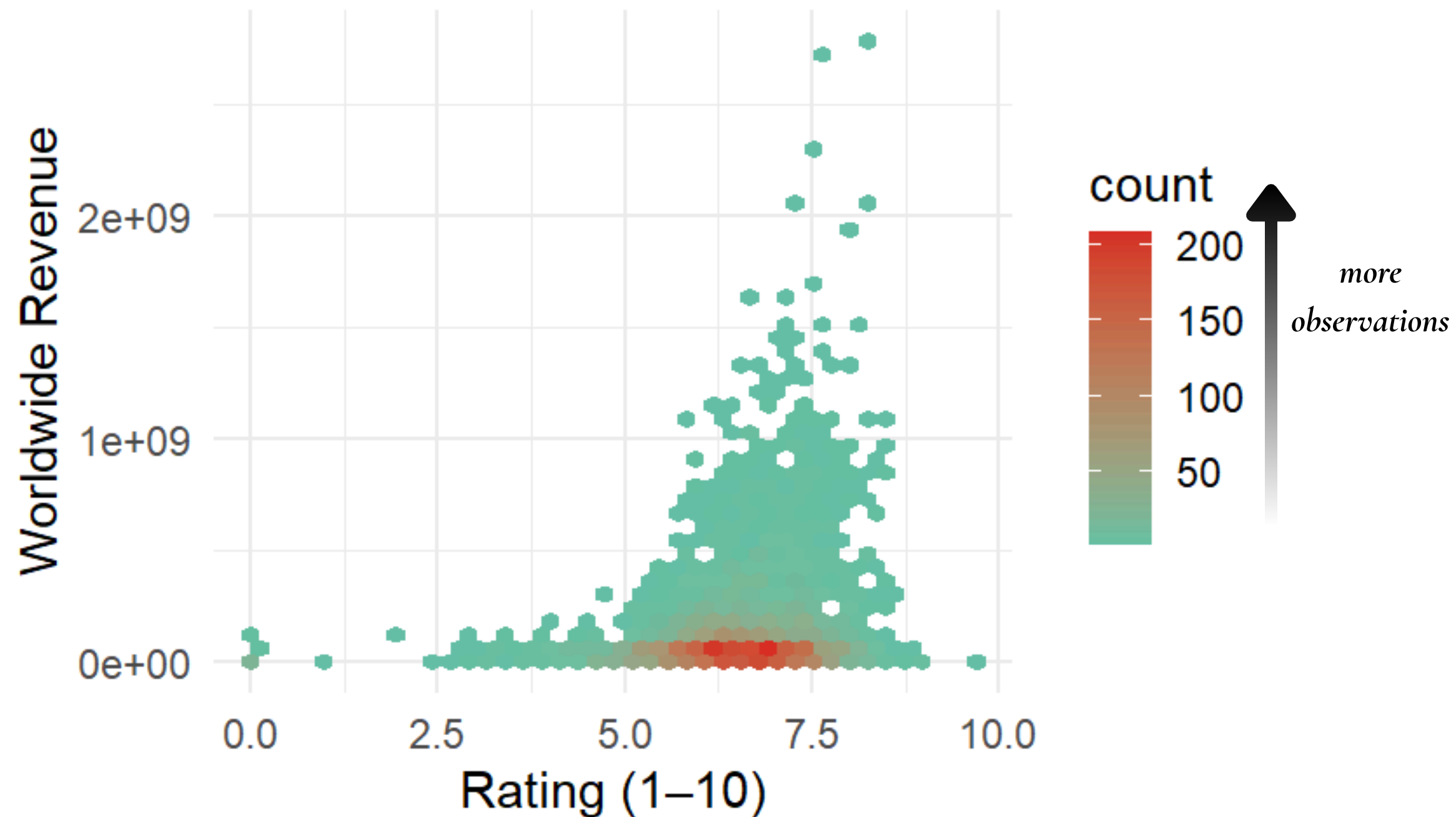
- As ratings increase, worldwide revenue tends to rise.

Rating vs Revenue (Hexbin Density)

Low Ratings ≤ 5.9

Medium Ratings 6.0 – 7.4

High Ratings ≥ 7.5



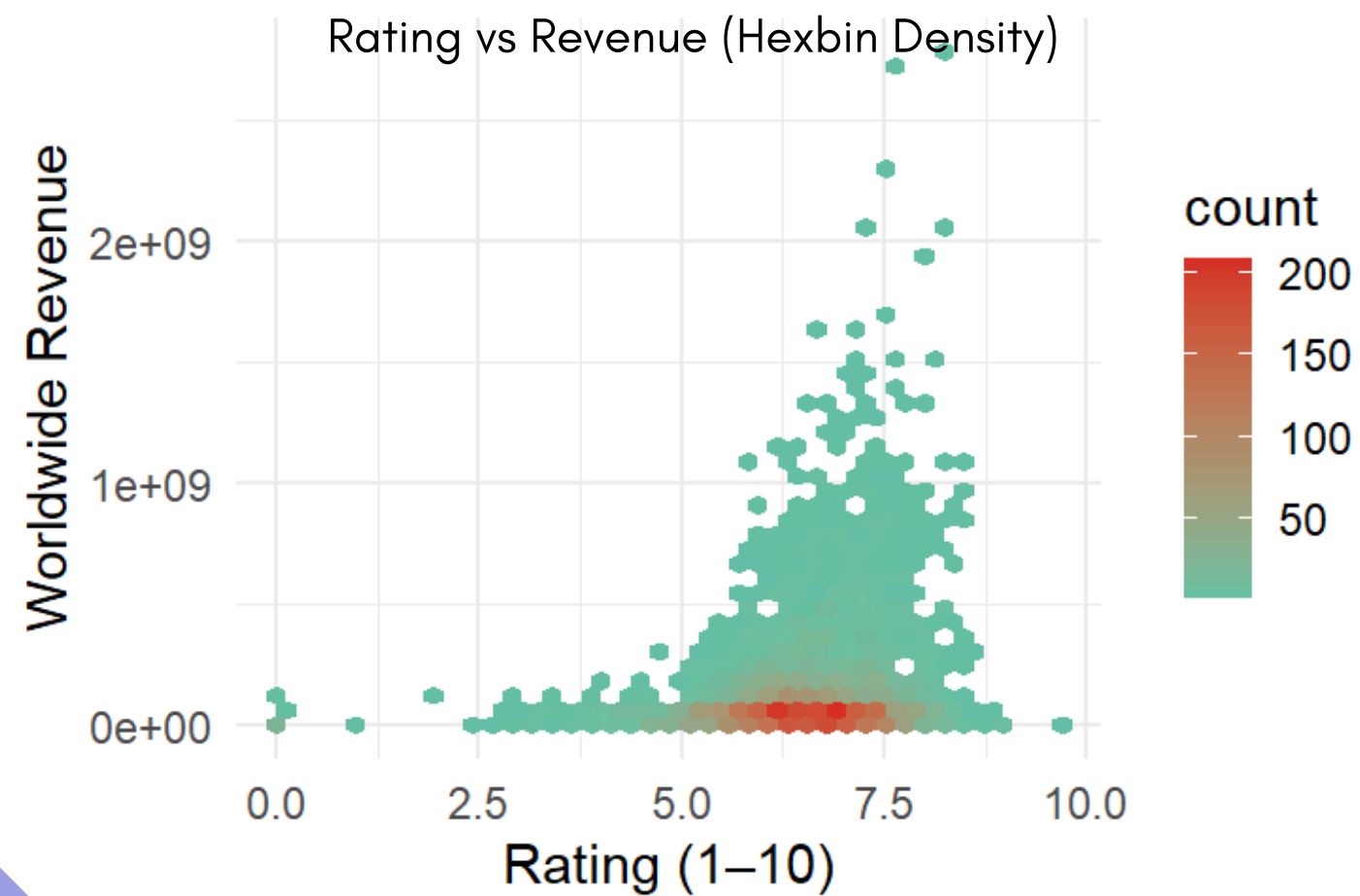
Higher feedback generally corresponds to higher performance, with most observations in the medium range.



Inferential analysis

making conclusions from data

Normality Check – Skewness Analysis



Hexbin Plot (Ratings vs Revenue):

- Most movies cluster at medium ratings with modest revenue, a few extreme blockbusters push the distribution.
- Revenue distribution is skewed.

Shapiro–Wilk Test Result:

- $W = 0.538$
- $p < 2.2e-16$

Revenue data is not normally distributed, so parametric tests like ANOVA are not suitable; non-parametric methods will be used instead.

Spearman's Correlation – Rating vs Revenue

- **H₀ (Null Hypothesis):** No monotonic relationship between rating and revenue ($\rho = 0$)
- **H₁ (Alternative Hypothesis):** There is a monotonic relationship between rating and revenue ($\rho \neq 0$)

- **Results:**

Spearman's rho (ρ) = 0.1405 ($\rho > 0 \rightarrow$ positive monotonic relationship)

p-value < 2.2e-16 $\rightarrow$ Highly significant

- **Conclusion**

Weak but positive relationship

Higher ratings $\rightarrow$ higher revenue

Non-Parametric Results

Kruskal–Wallis & Dunn's Test Hypothesis

H₀ (Null Hypothesis): Median revenues are equal across rating categories (Low, Medium, High).

H₁ (Alternative Hypothesis): At least one rating category differs in median revenue

Kruskal–Wallis Test

$$\chi^2 = 100.49, \text{ df} = 2, p < 0.001$$

Reject H₀ → Significant differences among groups. Therefore, at least one group's revenue distribution is different.

Dunn's Pairwise Comparisons

Low vs Medium: $z = 8.51$, $p < 0.001 \rightarrow$ significant.

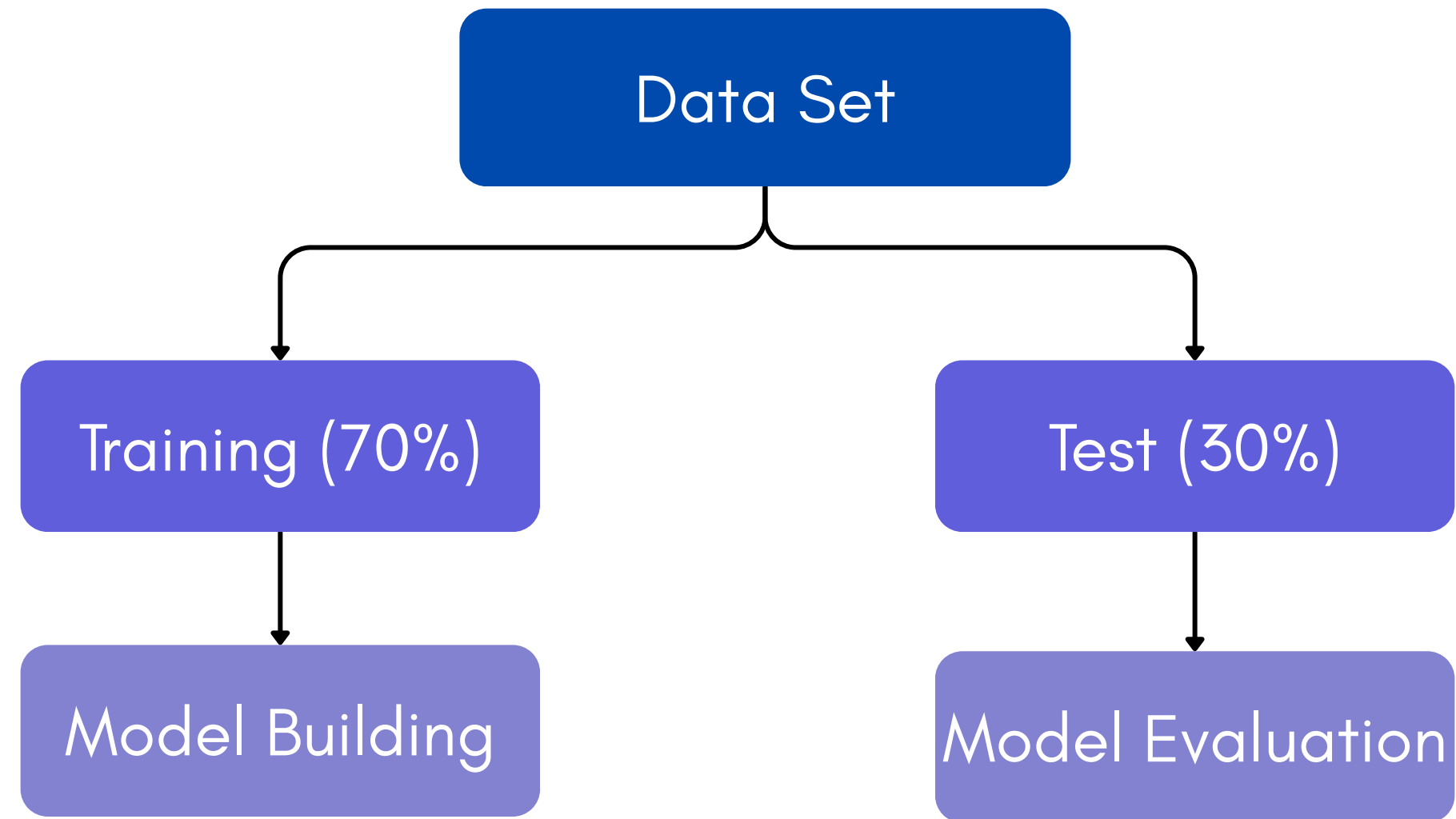
Medium vs High: $z = 3.46$, $p = 0.0008 \rightarrow$ significant.

Low vs High: $z = 9.10$, $p < 0.001 \rightarrow$ significant.

All rating categories differ significantly in revenue. Clear monotonic trend: $\text{Low} < \text{Medium} < \text{High}$.
Higher ratings consistently lead to higher revenue.



Predictive Analysis



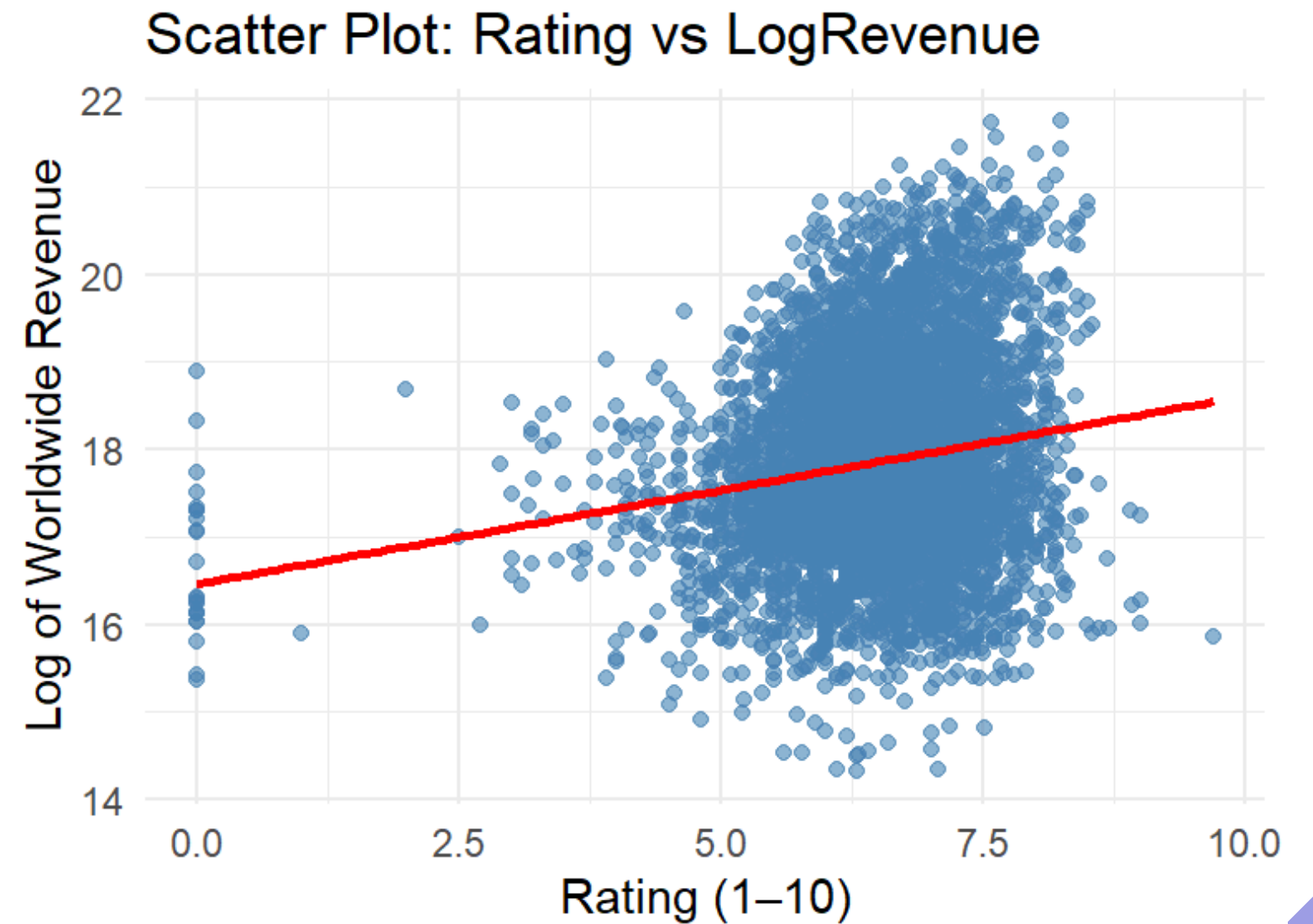
- Dataset split to avoid overfitting.
- Training set (70%) used to build the model.
- Testing set (30%) used to evaluate predictive accuracy.

Linear Regression Model

Equation Form

$$\text{LogRevenue} = 16.44 + 0.218 \cdot \text{Rating}$$

- Rating has a positive effect on revenue (significant).
- Each 1-point increase in rating $\rightarrow$ +0.22 in LogRevenue.
- Model explains ~2.8% of revenue variance (weak predictor).



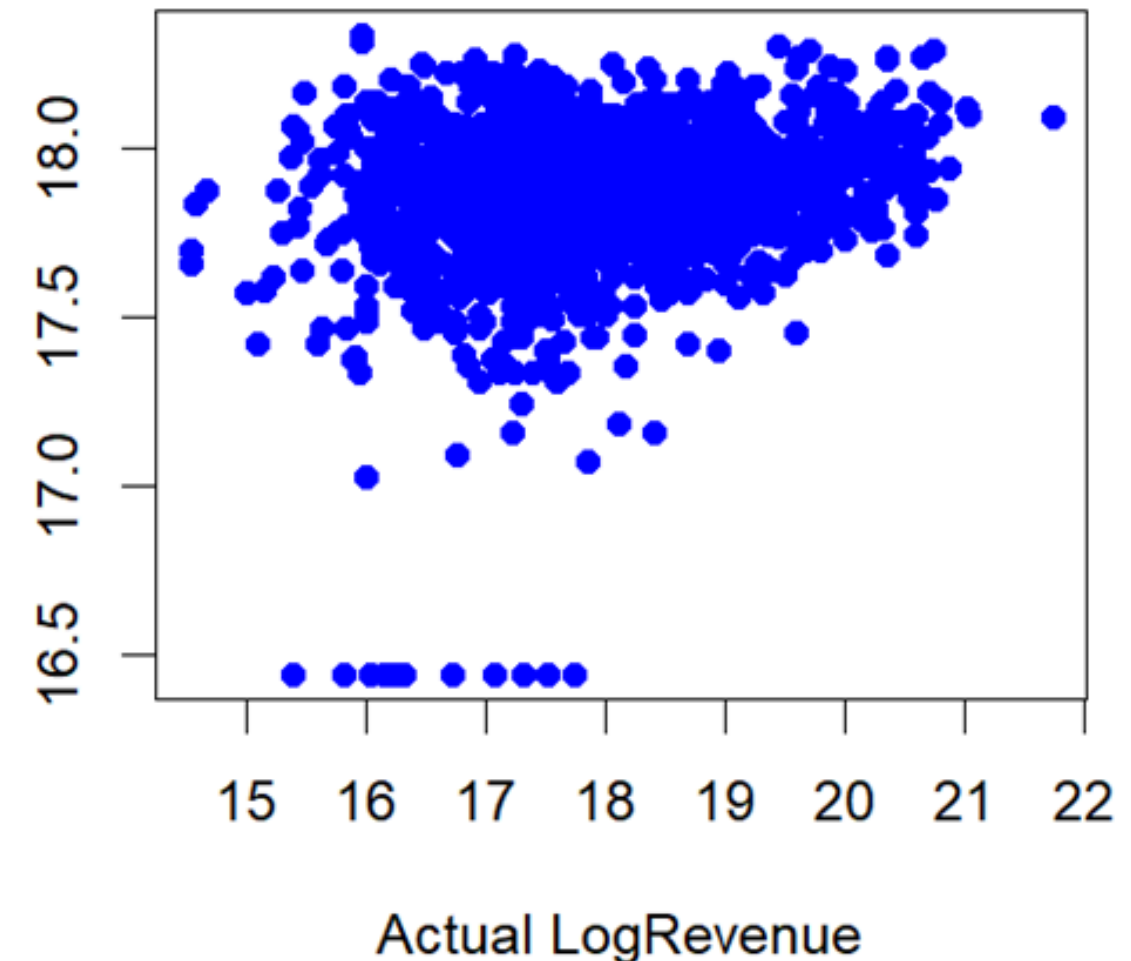
Model Evaluation – Rating → Revenue Results

$R^2 = 0.034 \rightarrow$ Rating explains only ~3% of revenue variance.

$RMSE = 1.14 \rightarrow$ Prediction error is moderate.

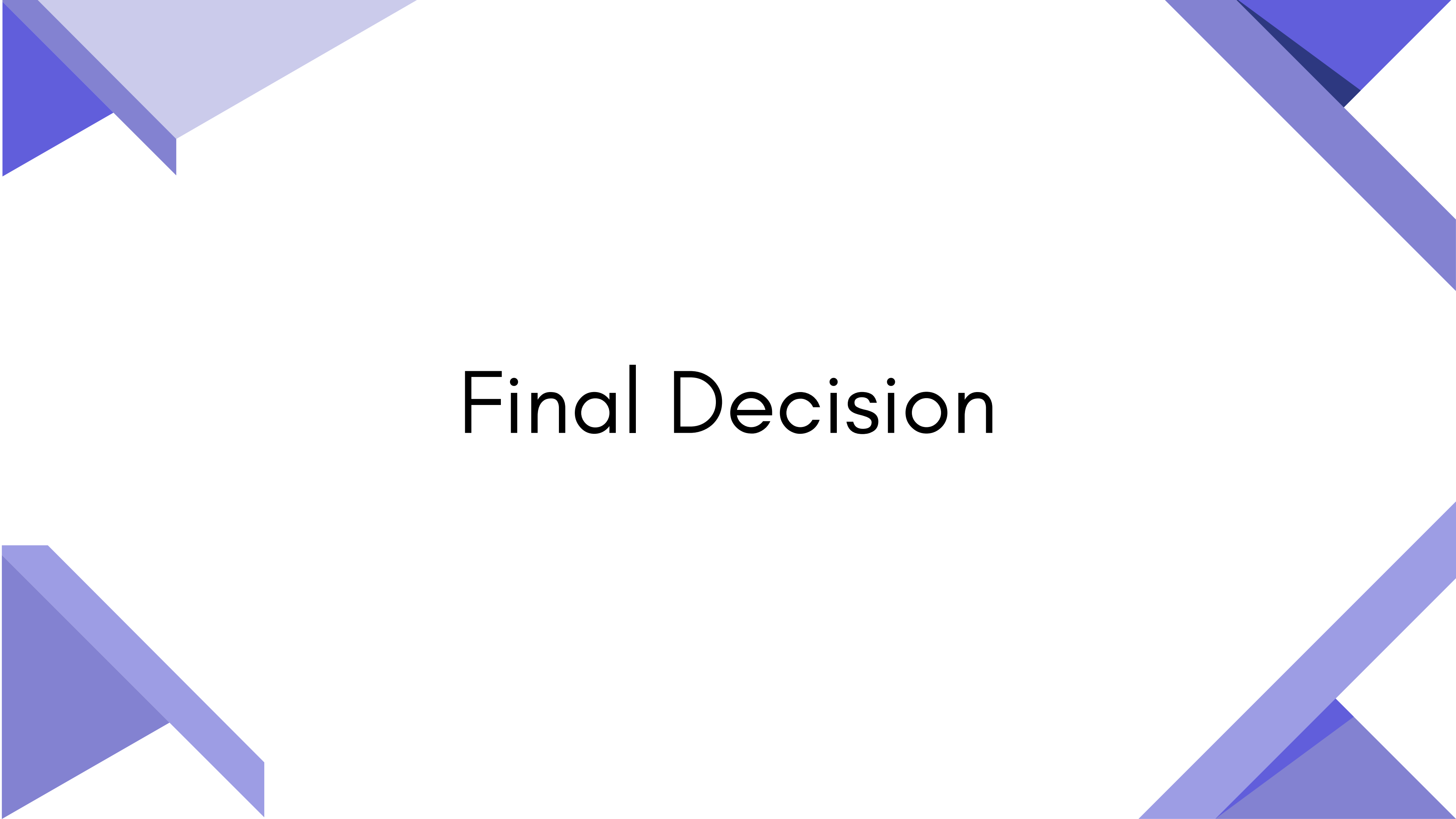
- Model is statistically significant but weak in predictive power.
- Predictions underestimate actual revenue.
- Rating alone cannot predict revenue well; other variables (budget, etc.) may be stronger predictors.

Predicted vs Actual Revenue

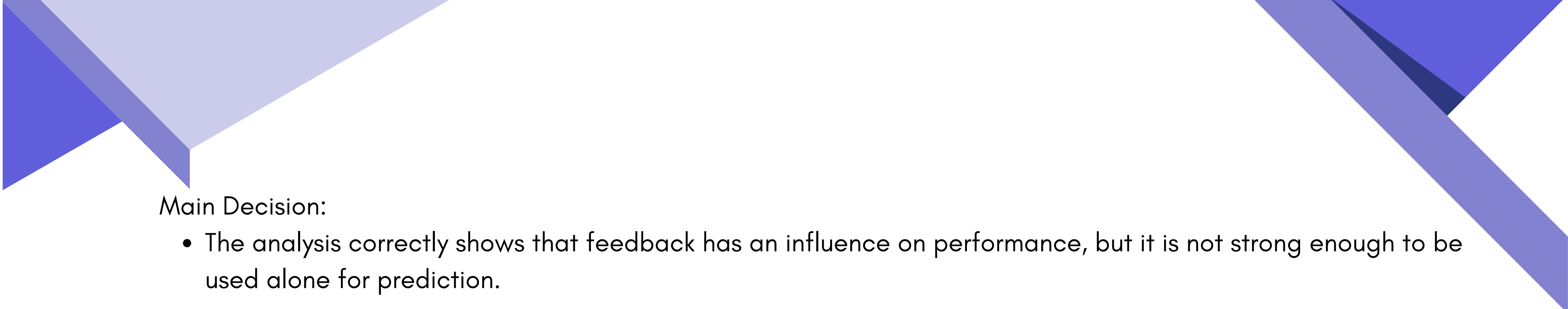


Conclusion – Predictive Analysis

- Rating has a positive but weak relationship with revenue.
- Model is statistically significant but explains little variance ($R^2 \approx 3\%$).
- Predictions consistently underestimate actual revenue.

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Final Decision



Main Decision:

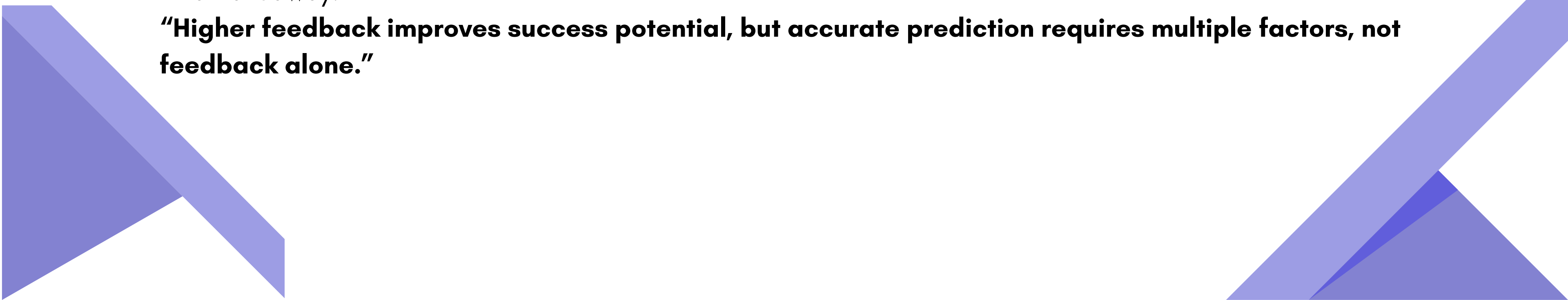
- The analysis correctly shows that feedback has an influence on performance, but it is not strong enough to be used alone for prediction.

Improvements:

- Include additional factors such as budget, category, and engagement metrics
- Focus on improving quality to achieve higher feedback levels
- Use multiple variables together for better evaluation
- Apply more advanced models for improved prediction accuracy

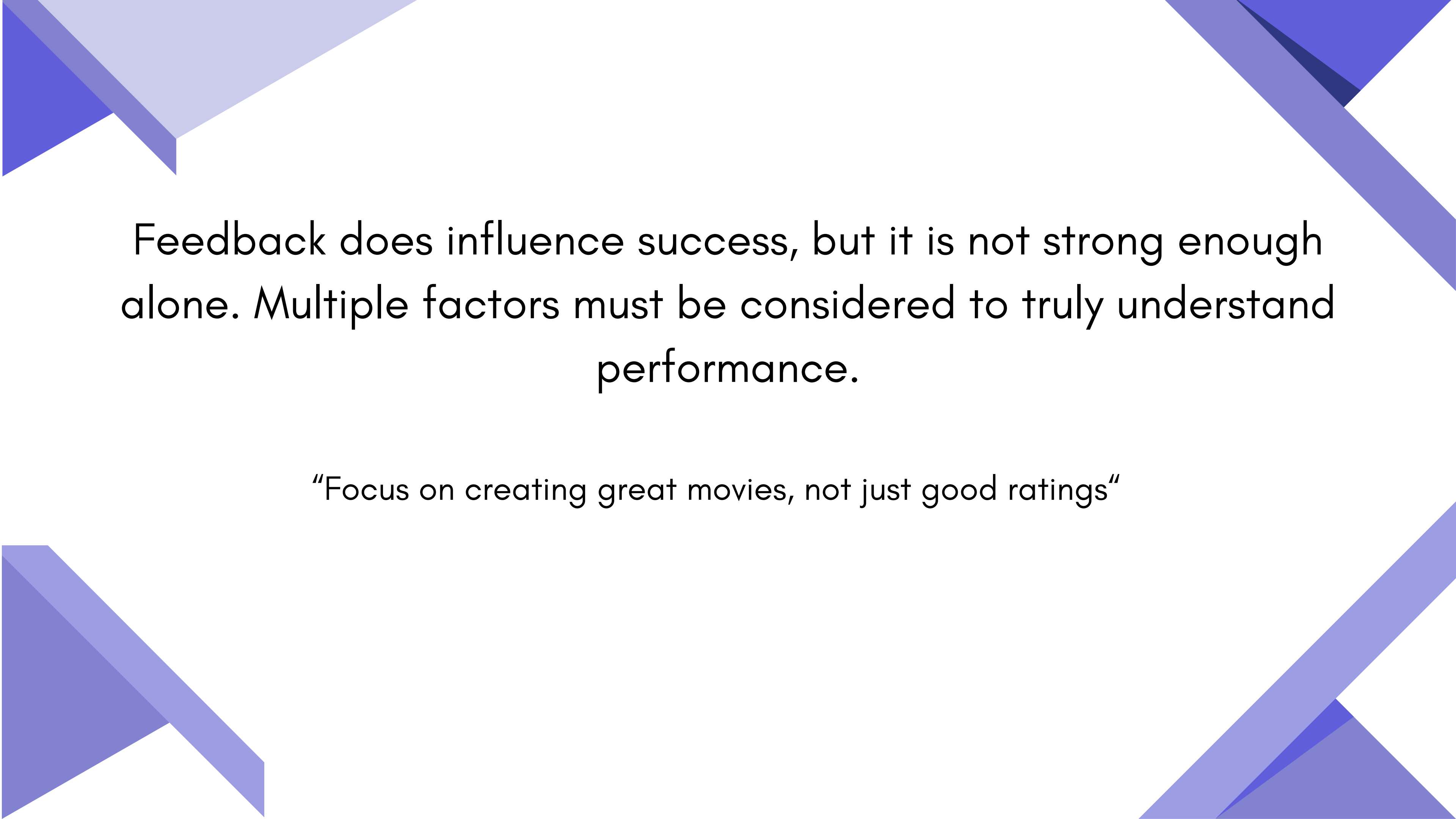
Final Takeaway:

“Higher feedback improves success potential, but accurate prediction requires multiple factors, not feedback alone.”



The background features four decorative geometric shapes in the corners, each composed of overlapping triangles in various shades of blue and purple. The top-left shape is a light purple triangle with a darker blue triangle on top. The top-right shape is a dark blue triangle with a medium blue triangle on top. The bottom-left shape is a medium blue triangle with a light purple triangle on top. The bottom-right shape is a light purple triangle with a dark blue triangle on top.

Conclusion



Feedback does influence success, but it is not strong enough alone. Multiple factors must be considered to truly understand performance.

“Focus on creating great movies, not just good ratings”

The image features a decorative graphic in the top-left corner consisting of several overlapping triangular and quadrilateral shapes in various shades of purple, ranging from a deep indigo to a light lavender. These shapes create a layered, geometric effect. The rest of the image is a solid, light cream or off-white background.

Thank You